

BS-PH101 — Physics-I

Semester I/II • 3L:1T:0P • 4 Credits

Module 4 Comprehensive Exam Notes: Quantum Mechanics

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Inadequacy of Classical Physics and Black Body Radiation

1.1 Failure of Classical Mechanics and Planck's Radiation Law

Classical physics (Rayleigh-Jeans law $u(\nu)d\nu = \frac{8\pi\nu^2}{c^3}k_B T d\nu$) predicted that black body radiation energy density tends to infinity at short wavelengths ($\lambda \rightarrow 0$), known as the **Ultraviolet Catastrophe**.

Planck's Quantum Hypothesis (1900)

Black body cavity walls consist of atomic oscillators that absorb or emit electromagnetic radiation in discrete packets called **quanta** (photons) of energy:

$$E = nh\nu \quad (n = 1, 2, 3, \dots) \quad (1)$$

where $h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$ is Planck's constant.

Planck's Black Body Radiation Law Derivation

Average energy of an oscillator $\langle E \rangle = \frac{\sum_{n=0}^{\infty} nh\nu e^{-nh\nu/k_B T}}{\sum_{n=0}^{\infty} e^{-nh\nu/k_B T}} = \frac{h\nu}{e^{h\nu/k_B T} - 1}$. Multiplying by number of modes per unit volume $N(\nu)d\nu = \frac{8\pi\nu^2}{c^3}d\nu$:

$$u(\nu)d\nu = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/k_B T} - 1} d\nu \iff u(\lambda)d\lambda = \frac{8\pi hc}{\lambda^5} \frac{1}{e^{hc/\lambda k_B T} - 1} d\lambda \quad (2)$$

Deduction of Limiting Laws:

- **Rayleigh-Jeans Law (Long λ):** For $h\nu \ll k_B T$, $e^{h\nu/k_B T} - 1 \approx \frac{h\nu}{k_B T} \implies u(\nu)d\nu = \frac{8\pi\nu^2}{c^3}k_B T d\nu$.
- **Wien's Distribution Law (Short λ):** For $h\nu \gg k_B T$, $e^{h\nu/k_B T} - 1 \approx e^{h\nu/k_B T} \implies u(\lambda)d\lambda = \frac{8\pi hc}{\lambda^5} e^{-hc/\lambda k_B T} d\lambda$.

2. Compton Effect, de Broglie Hypothesis, and Matter Waves

2.1 Compton Effect

When a high-energy X-ray or γ -ray photon collides elastically with a stationary free electron, the scattered photon emerges with a longer wavelength ($\lambda' > \lambda$).

Derivation of Compton Shift Formula

Applying conservation of relativistic energy and momentum during collision:

$$\text{Energy Conservation: } h\nu + m_0 c^2 = h\nu' + mc^2 \quad (3)$$

$$\text{Momentum Conservation: } \vec{p}_{\text{photon}} = \vec{p}'_{\text{photon}} + \vec{p}_{\text{electron}} \quad (4)$$

Eliminating electron velocity v using $m = \frac{m_0}{\sqrt{1-v^2/c^2}}$ yields the **Compton Shift Equation**:

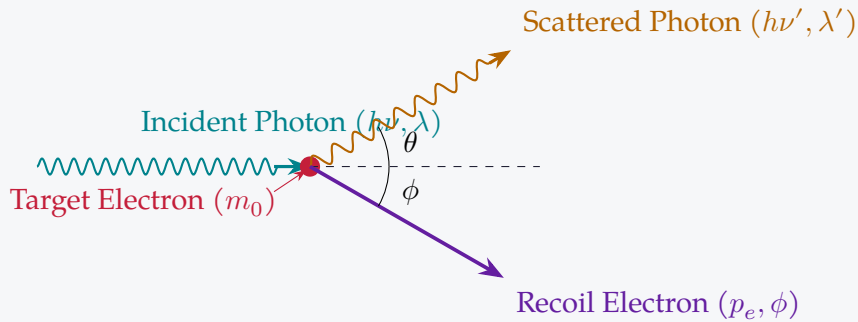
$$\Delta\lambda = \lambda' - \lambda = \frac{h}{m_0 c} (1 - \cos \theta) = 2\lambda_c \sin^2 \left(\frac{\theta}{2} \right) \quad (5)$$

where $\lambda_c = \frac{h}{m_0 c} = 0.02426 \text{ \AA} = 2.426 \times 10^{-12} \text{ m}$ is the **Compton Wavelength**.

Key Characteristics of Compton Shift:

- $\Delta\lambda$ depends *only* on scattering angle θ , independent of incident wavelength or target material.
- Maximum shift occurs at $\theta = 180^\circ$ (backscattering): $\Delta\lambda_{\max} = \frac{2h}{m_0 c} = 0.0485 \text{ \AA}$.
- Compton shift is absent for visible light because $\Delta\lambda \ll \lambda_{\text{visible}}$ is unresolvable.

Compton Scattering Geometry



2.2 de Broglie Hypothesis and Davisson-Germer Experiment

Definition: de Broglie Hypothesis

Louis de Broglie (1924) proposed that moving matter particles exhibit wave properties with a wavelength (λ) inversely proportional to momentum (p):

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{2mqV}} \quad (6)$$

For an electron accelerated through potential V volts:

$$\lambda_e = \frac{12.27}{\sqrt{V}} \text{ \AA} \quad (7)$$

Davisson-Germer Experiment: Accelerated electrons at $V = 54 \text{ V}$ onto a Nickel crystal target observed a pronounced scattering peak at angle $\phi = 50^\circ$. Bragg diffraction angle $\theta = 90^\circ - \phi/2 = 65^\circ$.

$$\lambda_{\text{Bragg}} = 2d \sin \theta = 2(0.91 \text{ \AA}) \sin 65^\circ = 1.65 \text{ \AA} \quad \text{vs} \quad \lambda_{\text{de Broglie}} = \frac{12.27}{\sqrt{54}} = 1.67 \text{ \AA} \quad (8)$$

This excellent agreement provided direct experimental verification of matter waves.

Solved Problem 1: Compton Shift at 90 Degrees

Problem: X-rays of wavelength $\lambda = 1.0 \text{ \AA}$ are scattered by a target at $\theta = 90^\circ$. Calculate: (a) Scattered wavelength λ' , (b) Energy of recoil electron.

Solution:

$$1. \Delta\lambda = \lambda_c(1 - \cos 90^\circ) = 0.02426 \text{ \AA}(1 - 0) = 0.02426 \text{ \AA}.$$

$$2. \text{Scattered wavelength } \lambda' = \lambda + \Delta\lambda = 1.0 + 0.02426 = \mathbf{1.02426 \text{ \AA}}.$$

$$3. \text{Recoil electron kinetic energy } K_e = hc \left(\frac{1}{\lambda} - \frac{1}{\lambda'} \right) = hc \frac{\Delta\lambda}{\lambda\lambda'}:$$

$$K_e = \frac{(6.626 \times 10^{-34})(3 \times 10^8)(0.02426 \times 10^{-10})}{(1.0 \times 10^{-10})(1.02426 \times 10^{-10})} = 4.706 \times 10^{-17} \text{ J} = \mathbf{294.1 \text{ eV}} \quad (9)$$

3. Heisenberg Uncertainty Principle

3.1 Uncertainty Relations

Heisenberg Uncertainty Principle Relations

It is impossible to simultaneously measure both the exact position and exact momentum of a particle with absolute precision:

$$\Delta x \cdot \Delta p_x \geq \frac{\hbar}{2} \quad \left(\text{where } \hbar = \frac{h}{2\pi} = 1.054 \times 10^{-34} \text{ J} \cdot \text{s} \right) \quad (10)$$

Alternate canonical conjugate forms:

$$\text{Energy-Time Uncertainty: } \Delta E \cdot \Delta t \geq \frac{\hbar}{2} \quad (11)$$

$$\text{Angular Position-Momentum: } \Delta\theta \cdot \Delta L_z \geq \frac{\hbar}{2} \quad (12)$$

3.2 Applications: Non-Existence of Electron in Nucleus

Proof: Non-Existence of Electron in Nucleus

Premise: Radius of atomic nucleus is $r \approx 5 \times 10^{-15} \text{ m}$. If an electron exists inside, maximum position uncertainty $\Delta x \approx 10^{-14} \text{ m}$.

1. Minimum momentum uncertainty:

$$\Delta p \geq \frac{\hbar}{2\Delta x} = \frac{1.054 \times 10^{-34}}{2 \times 10^{-14}} = 5.27 \times 10^{-21} \text{ kg} \cdot \text{m/s} \quad (13)$$

2. Minimum relativistic energy of electron ($E \approx pc$ for $p \gg m_0c$):

$$E_{\min} \approx \Delta p \cdot c = (5.27 \times 10^{-21}) \times (3 \times 10^8) = 1.58 \times 10^{-12} \text{ J} \approx \mathbf{9.9 \text{ MeV}} \quad (14)$$

3. Experimental β -decay electrons emitted from nucleus have maximum energies of only 2 – 3 MeV.

Conclusion: Electrons **cannot reside** inside the nucleus.

4. Schrödinger Wave Equation and Wave Function

4.1 Physical Interpretation of Wave Function ψ

The wave function $\psi(\vec{r}, t)$ describes the quantum state of a particle.

Born's Probability Interpretation of Wave Function

ψ itself is a complex quantity without direct physical measurement. However, the scalar product $|\psi(\vec{r}, t)|^2 = \psi^* \psi$ represents the **probability density** of finding the particle at position $\vec{r}$ at time t .

$$P = \int_{-\infty}^{\infty} |\psi(x, t)|^2 dx = 1 \quad (\text{Normalization Condition}) \quad (15)$$

Criteria for an Acceptable Quantum Wave Function:

1. ψ must be **single-valued** at all points in space.
2. ψ and its spatial derivative $\frac{\partial \psi}{\partial x}$ must be **continuous** everywhere.
3. ψ must be **square-integrable** (normalizable, $\psi \rightarrow 0$ as $r \rightarrow \infty$).

4.2 Time-Dependent and Time-Independent Schrödinger Equations

Schrödinger Wave Equations

• 1D Time-Dependent Schrödinger Equation:

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x, t)}{\partial x^2} + V(x, t) \psi(x, t) = i\hbar \frac{\partial \psi(x, t)}{\partial t} \quad (16)$$

• 1D Time-Independent Schrödinger Equation (for stationary states $\psi(x, t) = \psi(x)e^{-iEt/\hbar}$):

$$\frac{d^2 \psi(x)}{dx^2} + \frac{2m}{\hbar^2} (E - V(x)) \psi(x) = 0 \iff \hat{H} \psi = E \psi \quad (17)$$

where $\hat{H} = -\frac{\hbar^2}{2m} \nabla^2 + V$ is the Hamiltonian operator.

5. Quantum Applications: Particle in Box, Harmonic Oscillator, H-Atom

5.1 Particle in a 1D Infinite Potential Box

Consider a particle of mass m confined in a 1D box of length L ($V(x) = 0$ for $0 < x < L$, and $V(x) = \infty$ outside).

Inside the box ($0 < x < L$), $\frac{d^2\psi}{dx^2} + k^2\psi = 0$ where $k = \frac{\sqrt{2mE}}{\hbar}$. Boundary conditions $\psi(0) = 0$ and $\psi(L) = 0$ require $kL = n\pi$.

1D Box Eigenvalues and Normalised Eigenfunctions

$$\text{Energy Eigenvalues: } E_n = \frac{n^2\pi^2\hbar^2}{2mL^2} = \frac{n^2\hbar^2}{8mL^2} \quad (n = 1, 2, 3, \dots) \quad (18)$$

$$\text{Normalised Wavefunctions: } \psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \quad (19)$$

Key Consequences:

- **Energy Quantization:** Energy levels are discrete and proportional to n^2 .
- **Zero-Point Energy (E_1):** The lowest possible energy is $E_1 = \frac{\hbar^2}{8mL^2} \neq 0$, agreeing with uncertainty principle.
- **Degeneracy in 3D Cubic Box:** $E_{n_x, n_y, n_z} = \frac{\hbar^2}{8mL^2}(n_x^2 + n_y^2 + n_z^2)$. States with different quantum numbers sharing same energy are **degenerate** (e.g. $(1, 1, 2), (1, 2, 1), (2, 1, 1)$ is 3-fold degenerate).

5.2 Quantum Harmonic Oscillator and Hydrogen Atom

Quantum Harmonic Oscillator and Hydrogen Atom Energies

- **Quantum Harmonic Oscillator ($V = \frac{1}{2}m\omega^2x^2$):**

$$E_n = \left(n + \frac{1}{2}\right) \hbar\omega \quad (n = 0, 1, 2, \dots) \quad (20)$$

Zero-point energy $E_0 = \frac{1}{2}\hbar\omega$ (unlike classical oscillator which can rest at zero energy).

- **Hydrogen Atom (Coulomb Potential $V = -\frac{e^2}{4\pi\epsilon_0 r}$):**

$$E_n = -\frac{me^4}{32\pi^2\epsilon_0^2\hbar^2n^2} = -\frac{13.6}{n^2} \text{ eV} \quad (n = 1, 2, 3, \dots) \quad (21)$$

Described by four quantum numbers: Principal (n), Orbital (l), Magnetic (m_l), and Spin (m_s).

Solved Problem 2: Particle in 1D Box Transition Energy

Problem: An electron is trapped in a 1D infinite box of width $L = 1.0 \text{ \AA}$. Calculate: (a) Ground state energy E_1 in eV, (b) Wavelength of photon emitted during transition from $n = 2 \rightarrow n = 1$.

Solution:

1. Ground state energy $E_1 = \frac{h^2}{8mL^2}$:

$$E_1 = \frac{(6.626 \times 10^{-34})^2}{8 \times (9.11 \times 10^{-31}) \times (1.0 \times 10^{-10})^2} = 6.025 \times 10^{-18} \text{ J} = \mathbf{37.66 \text{ eV}} \quad (22)$$

2. First excited energy $E_2 = 2^2 E_1 = 4(37.66) = 150.64 \text{ eV}$.
3. Energy difference $\Delta E = E_2 - E_1 = 3E_1 = 112.98 \text{ eV} = 1.808 \times 10^{-17} \text{ J}$.
4. Emitted wavelength $\lambda = \frac{hc}{\Delta E} = \frac{(6.626 \times 10^{-34})(3 \times 10^8)}{1.808 \times 10^{-17}} = 1.10 \times 10^{-8} \text{ m} = \mathbf{110 \text{ \AA}}$.

Quick Revision — Module 4

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Key Formulas and Metrics at a Glance

Planck Photon Energy	$E = h\nu = \frac{hc}{\lambda}$
Planck Radiation Law	$u(\lambda)d\lambda = \frac{8\pi hc}{\lambda^5} \frac{1}{e^{hc/\lambda k_B T} - 1} d\lambda$
Compton Shift Equation	$\Delta\lambda = \lambda' - \lambda = \frac{h}{m_0 c} (1 - \cos\theta)$
Compton Wavelength	$\lambda_c = \frac{h}{m_0 c} = 0.02426 \text{ \AA}$
de Broglie Wavelength	$\lambda = \frac{h}{p} = \frac{12.27}{\sqrt{V}} \text{ \AA} \quad (\text{for electron})$
Heisenberg Uncertainty	$\Delta x \cdot \Delta p \geq \frac{\hbar}{2} \quad \text{and} \quad \Delta E \cdot \Delta t \geq \frac{\hbar}{2}$
1D Time-Indep. Schrödinger	$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V)\psi = 0$
1D Box Energy Eigenvalue	$E_n = \frac{n^2 h^2}{8mL^2} \quad (n = 1, 2, 3, \dots)$
1D Box Wavefunction	$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$
Harmonic Oscillator Energy	$E_n = (n + 1/2)\hbar\omega \quad (\text{Zero-point } E_0 = \frac{1}{2}\hbar\omega)$
Hydrogen Atom Energy	$E_n = -\frac{13.6}{n^2} \text{ eV}$

One-Line Definitions

- **Compton Effect:** The increase in wavelength of X-ray/ γ -ray photons when scattered by weakly bound target electrons.
- **Wave-Particle Duality:** The property of matter and light exhibiting both wave-like and particle-like behaviors depending on experimental setup.
- **Probability Density ($|\psi|^2$):** The probability per unit volume of finding a quantum particle at a specific spatial coordinate.
- **Zero-Point Energy:** The lowest possible non-zero ground-state energy that a quantum mechanical physical system may possess ($E_1 \neq 0$).
- **Degeneracy:** The phenomenon where two or more distinct quantum states share the exact same energy eigenvalue.

Frequently Asked / Viva Questions

Q1. What was Rayleigh-Jeans law and why did it fail?

Rayleigh-Jeans law ($u(\nu) \propto \nu^2 T$) predicted infinite energy density at high frequencies ($\lambda \rightarrow 0$), causing the ultraviolet catastrophe.

Q2. State Planck's quantum hypothesis.

Frequently Asked / Viva Questions (contd.)

Atomic oscillators absorb or emit electromagnetic radiation only in discrete energy quanta $E = nh\nu$.

Q3. Why does Compton effect observe no shift at $\theta = 0^\circ$ and maximum shift at $\theta = 180^\circ$?

$\Delta\lambda = \lambda_c(1 - \cos\theta)$. At $\theta = 0^\circ$, $\cos 0^\circ = 1 \Rightarrow \Delta\lambda = 0$. At $\theta = 180^\circ$, $\cos 180^\circ = -1 \Rightarrow \Delta\lambda = 2\lambda_c = 0.0485 \text{ \AA}$.

Q4. Why is Compton effect not observable with visible light?

Visible light has wavelength $\sim 5000 \text{ \AA}$. The maximum Compton shift ($0.0485 \text{ \AA}$) is less than 0.001% of λ_{visible} and cannot be resolved.

Q5. What is de Broglie wavelength of an electron accelerated through V volts?

$$\lambda = \frac{h}{\sqrt{2mqV}} = \frac{12.27}{\sqrt{V}} \text{ \AA}.$$

Q6. Explain the physical significance of Davisson-Germer experiment.

It demonstrated electron diffraction from a Nickel crystal lattice, proving experimentally that moving material particles possess wave characteristics.

Q7. Prove using uncertainty principle that electron cannot reside inside atomic nucleus.

For nuclear radius $\Delta x \approx 10^{-14} \text{ m}$, $\Delta p \geq \hbar/2\Delta x \Rightarrow E \approx \Delta p \cdot c \approx 9.9 \text{ MeV}$. Experimental β -decay energies are only $2 - 3 \text{ MeV}$, proving electrons cannot reside in nucleus.

Q8. State Born's physical interpretation of wave function ψ .

ψ itself has no direct physical measurement, but $|\psi(\vec{r}, t)|^2 = \psi^*\psi$ represents the probability density of finding the particle at $\vec{r}$ at time t .

Q9. What are the mandatory conditions for an acceptable wave function?

ψ must be single-valued, continuous everywhere along with its first spatial derivative, and square-integrable ($\int |\psi|^2 dV = 1$).

Q10. Why is ground state energy of particle in 1D box non-zero?

If $E_1 = 0 \Rightarrow p = 0 \Rightarrow \Delta p = 0$, requiring $\Delta x = \infty$ by uncertainty principle, contradicting confinement in box of finite length L .

Q11. What is zero-point energy of a quantum harmonic oscillator?

$E_0 = \frac{1}{2}\hbar\omega$. Unlike classical oscillators, quantum oscillators retain non-zero vibrational energy even at absolute zero temperature ($T = 0 \text{ K}$).

Q12. What is meant by degeneracy of energy levels?

Degeneracy occurs when multiple independent quantum wavefunctions (different sets of quantum numbers) correspond to the exact same energy eigenvalue (e.g. $(1, 1, 2)$, $(1, 2, 1)$, $(2, 1, 1)$ in a 3D box).

Summary Comparison of Quantum Models

System Model	Potential $V(x)$	Ground State Energy (E_1 or E_0)	Energy Spacing (ΔE)
Particle in 1D Box	$V = 0$ ($0 < x < L$), $V = \infty$ outside	$E_1 = \frac{h^2}{8mL^2}$	Non-uniform ($\propto n^2$).
Harmonic Oscillator	$V = \frac{1}{2}m\omega^2 x^2$	$E_0 = \frac{1}{2}\hbar\omega$	Uniform ($\Delta E = \hbar\omega$).
Hydrogen Atom	$V = -\frac{e^2}{4\pi\epsilon_0 r}$	$E_1 = -13.6 \text{ eV}$	Concentrated near continuum ($\propto -1/n^2$).